DensityKV: Density-Guided KV Cache Compression for Long Video Generation

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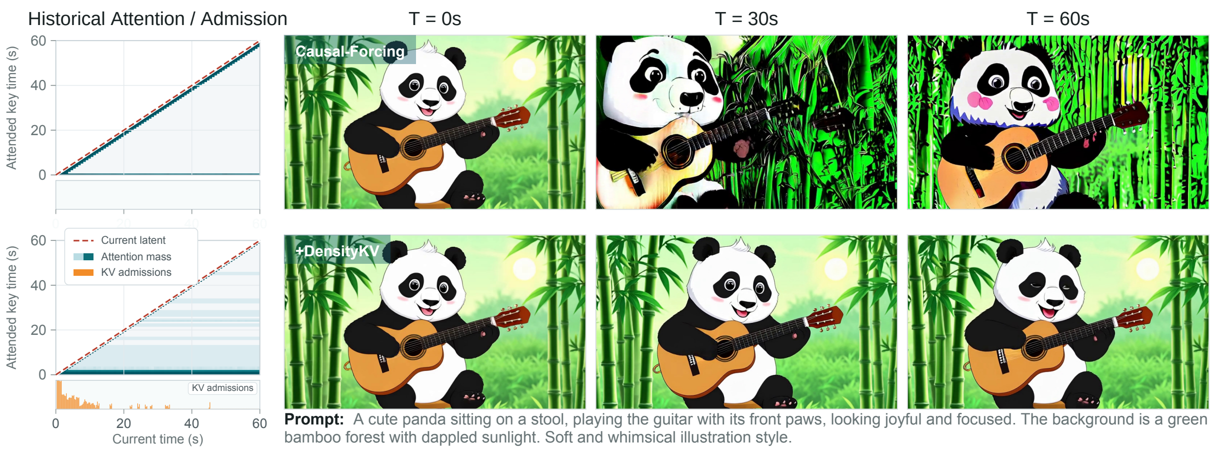

Figure 1: **DensityKV** maintains generated history in per-head KV banks whose local density growth is controlled after admission. Left: historical-attention maps show which past states each new segment uses, while admission profiles show when retained KV tokens entered the bank. Right: access to this retained history helps long videos maintain subjects, objects, and scenes more consistently over time.

Abstract

Autoregressive video diffusion models enable streaming generation through sliding-window attention, but each generated block is conditioned on previously generated content, causing appearance and motion errors to propagate recursively over time. Historical key–value (KV) memory preserves earlier subject and scene states and helps maintain long-horizon consistency. However, retaining every generated state creates a historical archive that grows continuously with the rollout, while recurrent states repeatedly add redundant coverage. To address this problem, we propose *DensityKV*, a training-free historical KV bank management strategy. DensityKV maintains a separate token-level KV bank for each attention head and measures local redundancy among the post-RoPE keys that directly parameterize attention routing using *Soft-Riesz density*. By constraining neighborhood-density growth after states enter the bank, DensityKV limits repeated historical accumulation while preserving coherent states from each completed generation block. Experiments across three autoregressive video generation backbones and multiple generation lengths show that, at the same upper bound on historical KV capacity, DensityKV improves long-horizon consistency and generation stability while keeping persistent historical storage bounded independently of rollout length.

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1 Introduction

Diffusion Transformers have become a strong foundation for video synthesis (Peebles and Xie 2023; Wan Team et al. 2025). Recent autoregressive (AR) variants turn short-clip diffusion models into causal generators that reuse key–value (KV) states and stream frames beyond the training horizon (Yin et al. 2024; Huang et al. 2025; Zhu et al. 2026; Yang et al. 2025). During long rollouts, recursive conditioning propagates appearance and motion errors, while the sliding window that bounds attention discards earlier subject and scene states that could stabilize later blocks.

Long-range memory restores access to this discarded history. Existing methods preserve early anchors, retain selected historical states, or retrieve non-local information from previously generated content (Yang et al. 2025; Yi et al. 2025; Zhao et al. 2026; Hu et al. 2026). However, full-history archives grow with every block and repeatedly store similar observations. We therefore ask: *before future queries are known, which exact historical K/V states should remain in a bounded bank?*

Frame-level retention is too coarse: a frame couples repetitive background tokens with rarer appearance, pose, and motion states. A shared keep/drop decision therefore either wastes capacity or removes useful evidence. Capacity should instead favor under-covered key neighborhoods. Because future queries are unknown, this allocation requires a query-independent criterion in the model's native attention space. This differs from global frame ranking: repeated regions should consume fewer slots, while evolving regions retain broader state coverage. Historical memory maintenance thus becomes a coverage problem over internal representations rather than a selection problem over complete frames.

Euclidean distance between post-RoPE keys bounds their normalized logit discrepancy for any query. Local key-space crowding can therefore measure redundant routing coverage without future queries, while values remain exactly paired with their original keys.

DensityKV is a training-free historical KV bank management strategy for frozen AR video generators. Each layer and head maintains a token-level bank of fully denoised states. A Soft-Riesz potential measures post-RoPE key crowding, while each state records its density at admission as a local baseline. DensityKV admits the largest candidate prefix satisfying the relative growth constraint and evicts states whose neighborhoods later become overrepresented. Co-admitted states establish their baselines together, so only later blocks contribute post-admission density growth. Native attention reads the resulting bank without a learned retriever or model fine-tuning. Figure 2 summarizes this online update.

Our contributions are:

- We formulate historical KV maintenance as an online representation-coverage problem and derive an attention-aligned post-RoPE key geometry for query-agnostic redundancy estimation.
- We introduce DensityKV, combining per-head token banks, Soft-Riesz crowding, and insertion-relative density constraints while preserving exact K/V pairs without training, merging, or reconstruction.
- Across three backbones and 30–120-second generations, DensityKV achieves the best overall average rank and 31 of 54 single-metric wins under the same upper bound on historical KV capacity.

2 Related Work

Long video generation. One line of work extends pre-trained bidirectional video diffusion models through tuning-free noise rescheduling or FIFO denoising (Qiu et al. 2023; Kim et al. 2024). A complementary line distills short-clip models into causal generators that produce clean frame blocks sequentially, enabling low-latency rollouts beyond the training horizon (Yin et al. 2024; Huang et al. 2025; Zhu et al. 2026; Yang et al. 2025). These approaches improve generation schedules, causal objectives, or rollout dynamics. Given a fixed AR backbone and local context, DensityKV instead selects the completed historical K/V states available to future blocks.

Memory for long video generation. Long-horizon video methods reintroduce discarded context in several ways. Persistent sinks and structured temporal buffers retain early anchors or selected historical frames (Yi et al. 2025; Zhao et al. 2026). Retrieval-based methods search generated history for non-local latent blocks, dynamic anchors, trusted references, or scene memories (Hu et al. 2026; Ye et al. 2026; Meng et al. 2026; Wu et al. 2026). More recent token-level policies estimate salience from head roles or future-query proxies, while latent-cache methods reduce the stored representation itself (Chen et al. 2026; Ji et al. 2026; Luo et al. 2026; Yesiltepe et al. 2026). These designs demonstrate that older generated states remain useful, but generally require a retrieval signal, a predicted query, or a transformed cache. DensityKV instead updates online as each clean block becomes available, preserving exact K/V pairs using only native post-RoPE key geometry. Unlike frame- or block-level selection, per-head token banks let regions with different rates of change compete separately for historical capacity.

Token-level KV bank management. KV-cache compression has been studied extensively for language models. Attention-driven methods preserve sinks, heavy hitters, or tokens assigned high head- and layer-specific importance (Xiao et al. 2023; Zhang et al. 2023; Li et al. 2024; Cai et al. 2024). Redundancy-aware methods further sample or cluster cached states, merge nearby representatives, or organize semantic clusters for query-conditioned recall (Zandieh et al. 2024; Cai et al. 2025; Hu et al. 2025; Liu et al. 2025). Autoregressive video presents a distinct online setting: every generated block adds dense spatiotemporal tokens, and repeatedly favoring the current query can make the memory follow an already drifting trajectory. DensityKV therefore maintains a query-agnostic subset independently in every attention head. It uses post-RoPE Euclidean geometry and a Soft-Riesz crowding potential (Hardin and Saff 2005), while frozen insertion baselines distinguish newly admitted support from density growth caused by later blocks.

3 Method

Problem Setup

Consider a frozen autoregressive video diffusion Transformer with L layers, H attention heads, and a local K/V window. At update t, a finalized, fully denoised block leaving the window supplies token-level candidates. Each candidate uses its native per-head post-RoPE key as both attention key and density descriptor and preserves its paired value. DensityKV retains at most M exact historical pairs per layer and head under an unbounded rollout and unknown future queries. Figure 2 outlines the geometry, density, and online update. Bold lowercase and calligraphic uppercase symbols denote vectors and collections, respectively.

Post-RoPE Attention Geometry

Let $\mathbf{q}, \mathbf{k} \in \mathbb{R}^{d_k}$ denote the actual per-head query and key vectors after the backbone applies RoPE. Their pre-softmax attention logit is $\lambda(\mathbf{q}, \mathbf{k}) = \mathbf{q}^\top \mathbf{k}/\sqrt{d_k}$. Cauchy–Schwarz yields

$$\sup_{\mathbf{q} \neq \mathbf{0}} \frac{|\lambda(\mathbf{q}, \mathbf{k}_i) - \lambda(\mathbf{q}, \mathbf{k}_j)|}{\|\mathbf{q}\|_2} = \frac{\|\mathbf{k}_i - \mathbf{k}_j\|_2}{\sqrt{d_k}}. \qquad (1)$$

Equality holds when $\mathbf{q} \parallel (\mathbf{k}_i - \mathbf{k}_j)$, so Equation (1) gives the exact worst-case normalized logit discrepancy, includ-

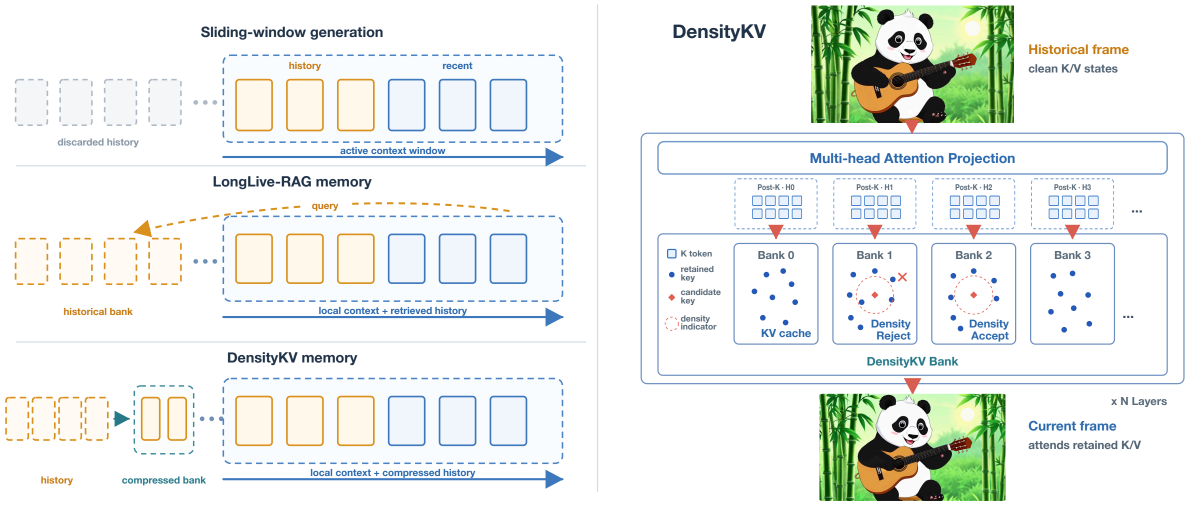

Figure 2: Overview of historical-memory construction and use. Left: sliding-window generation discards states outside the active context; LongLive-RAG stores discarded history and retrieves selected blocks; DensityKV instead maintains a bounded token-level history that is attended together with the local window. Right: at every Transformer layer, clean historical K/V states update independent per-head banks. Admission and rejection are determined by Soft-Riesz density over post-RoPE keys, while each value remains exactly paired with its original key. Current queries access the retained history through the backbone's native attention.

ing native RoPE. We use squared Euclidean distance because it preserves this neighborhood ordering, retains key-norm information absent from cosine distance, and supports batched evaluation. This geometry motivates a query-agnostic routing-coverage measure, not an attention-output error bound: nearby keys need not have interchangeable values. DensityKV reallocates finite capacity across key neighborhoods while copying or removing each selected K/V pair exactly. Derivations are provided in the supplementary material.

Soft-Riesz Density

Classical Riesz energy uses inverse-power interactions to penalize crowded point configurations (Hardin and Saff 2005). We regularize this interaction to obtain a finite, scale-aware kernel over the post-RoPE key geometry:

$$w(\mathbf{k}_i, \mathbf{k}_j) = \left(\epsilon + \frac{\|\mathbf{k}_i - \mathbf{k}_j\|_2^2}{\sigma^2}\right)^{-p}, \tag{2}$$

where σ sets neighborhood width, p controls decay, and ϵ keeps the interaction finite when two keys coincide. Since ϵ^{-p} is a common density scale and the effective bandwidth is $\sigma\sqrt{\epsilon}$, we fix $\epsilon = 1$ and use $(\sigma, p) = (8, 2)$ by default. For an indexed key multiset $\mathcal{A} = (\mathbf{k}_j)_{j=1}^{|\mathcal{A}|}$, we define the *Soft-Riesz density* of entry i as

$$\Phi_{\mathcal{A}}(i) = \sum_{\substack{j=1 \\ j \neq i}}^{|\mathcal{A}|} w(\mathbf{k}_i, \mathbf{k}_j). \tag{3}$$

High $\Phi_{\mathcal{A}}(i)$ indicates strong local support. DensityKV controls density growth *after admission*, rather than absolute density. The supplementary material gives the energy interpretation and calibration.

Online Density-Constrained Bank Update

Bank state and local baseline. At update t, layer ℓ and head h maintain

$$\mathcal{B}_t^{\ell,h} = \{(\bar{\mathbf{k}}_i^{\ell,h}, \bar{\mathbf{v}}_i^{\ell,h}, b_i^{\ell,h})\}_{i=1}^{m_t^\ell}, \qquad m_t^\ell \le M, \tag{4}$$

where $b_i^{\ell,h}$ is the insertion-density baseline and m_t^ℓ is the occupancy shared across heads through synchronized admission. The N arriving states form $\mathcal{C}_t^{\ell,h} = \{(\mathbf{k}_x^{\ell,h}, \mathbf{v}_x^{\ell,h})\}_{x=1}^N$. For any K/V collection $\mathcal{S}$, let $\mathcal{K}(\mathcal{S})$ denote its indexed post-RoPE key multiset, ignoring auxiliary fields such as b_i. Separate banks reflect head-specific projection geometries. We suppress t and ℓ when unambiguous.

Each admitted historical KV state $(\bar{\mathbf{k}}_i, \bar{\mathbf{v}}_i)$ stores a frozen insertion-density baseline for its post-RoPE key,

$$b_i = \max\left(\Phi_{\mathcal{K}_i^{\text{ins}}}(i), \delta\right), \tag{5}$$

where $\mathcal{K}_i^{\text{ins}}$ is the final key multiset produced by the update that admits state i, and $\delta > 0$ is a numerical floor. The frozen baseline measures subsequent growth relative to the neighborhood at admission.

Step 1: rank candidates by mean-normalized disturbance. For a nonempty bank, candidate x in head h receives

$$s_h(x) = \frac{1}{m_t} \sum_{i=1}^{m_t} \frac{w(\bar{\mathbf{k}}_i^h, \mathbf{k}_x^h)}{b_i^h}, \tag{6}$$

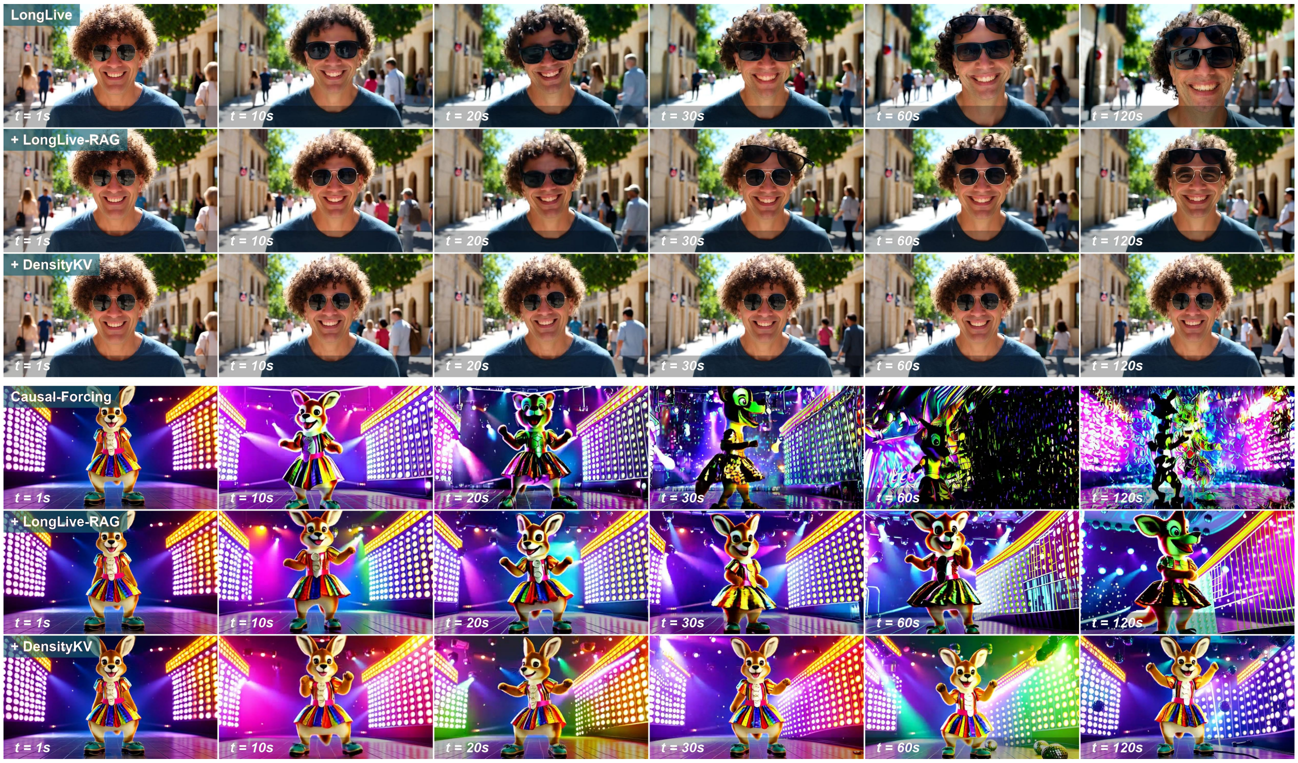

Figure 3: Qualitative comparison on two matched backbone–prompt pairs, each generated for 120 seconds. The upper group uses LongLive on MovieGenBench #097, the wig-and-sunglasses transformation; the lower group uses Causal-Forcing on MovieGenBench #012, the disco-dancing kangaroo.

which averages the normalized one-candidate increment over the bank. Each head sorts candidates by increasing s_h, producing permutation π_h and prefix

$$\mathcal{P}_r^h = \left\{ \left(\mathbf{k}_{\pi_h(j)}^h, \mathbf{v}_{\pi_h(j)}^h \right) \right\}_{j=1}^{r}. \tag{7}$$

We set $\mathcal{P}_0^h = \varnothing$. This step only determines the ordering; the next step evaluates cumulative prefix effects before admission.

Step 2: find the largest feasible cross-head prefix. If $\mathcal{P}_r^h$ were admitted, the projected density-growth ratio of retained state i would be

$$r_i^h(\mathcal{B}^h, \mathcal{P}_r^h) = \frac{\Phi_{\mathcal{K}(\mathcal{B}^h)}(i) + \sum_{j=1}^{r} w\left(\bar{\mathbf{k}}_i^h, \mathbf{k}_{\pi_h(j)}^h \right)}{b_i^h}. \tag{8}$$

The numerator combines current density with the prefix's cumulative contribution. Given a maximum growth factor $\tau > 1$, let

$$v_r^h = \left| \left\{ i : r_i^h(\mathcal{B}^h, \mathcal{P}_r^h) \geq \tau \right\} \right| \tag{9}$$

count the historical states whose projected ratio reaches τ. Separately, admitting r candidates requires

$$e_r = \max(0, m_t + r - M) \tag{10}$$

capacity-driven evictions. A prefix is feasible when $v_r^h \leq e_r$, so all violating historical states fit within the required eviction budget. The layer selects

$$r^\star = \max_{\substack{0 \leq r \leq \min(N,M) \\ v_r^h \leq e_r,\ \forall h}} r, \qquad r \in \mathbb{Z}. \tag{11}$$

The previous invariant makes $r = 0$ feasible, so the most restrictive head controls the shared cardinality; each head still uses its own $\mathcal{P}_{r^\star}^h$. The bank may remain below M when no positive prefix is feasible.

Step 3: evict, commit, and preserve the invariant. For each head, all retained states satisfying $r_i^h(\mathcal{B}^h, \mathcal{P}_{r^\star}^h) \geq \tau$ are marked for mandatory eviction. If there are fewer than $e_{r^\star}$ such states, DensityKV removes additional historical states in decreasing projected Soft-Riesz density until exactly $e_{r^\star}$ slots are released. With eviction set $\mathcal{R}^h$, it commits

$$\mathcal{B}_{t+1}^h = \left(\mathcal{B}_t^h \setminus \mathcal{R}^h \right) \cup \mathcal{P}_{r^\star}^h. \tag{12}$$

Since $w \geq 0$, evicting states cannot increase survivor density, so all surviving historical states satisfy the growth bound. New states derive b_i from the final bank: same-block candidate interactions become part of their frozen admission baselines, whereas later blocks contribute post-admission growth. Thus the feasibility test constrains a prefix's cumulative effect on existing history while preserving coherent same-block admission. Keys determine all decisions, and values move unchanged with their paired keys. An empty bank admits up to M source-ordered candidates and computes their baselines in the resulting bank. Pseudocode is provided in the supplementary material.

For T finalized tokens per head, full history requires $O(LHT(d_k + d_v))$ persistent K/V storage; DensityKV requires $O(LHM(d_k + d_v))$, independent of rollout length.

Time	Backbone	Method	Subject↑	Background↑	Smoothness↑	Dynamic↑	Aesthetic↑	Imaging↑	Avg. Rank↓
30s	Self-Forcing (2025)	Native	96.21	95.39	98.39	**52.03**	56.69	63.31	4.00
		+ ∞-RoPE (2025)	97.32	96.38	98.59	46.82	56.78	63.93	2.83
		+ Deep Forcing (2025)	97.04	96.02	98.57	38.85	56.44	61.91	4.50
		+ LongLive-RAG (2026)	97.57	96.56	**98.76**	42.24	57.17	65.43	2.17
		+ **DensityKV**	**97.79**	**96.72**	98.72	44.79	**57.68**	**66.93**	**1.50**
	LongLive (2025)	Native	97.35	96.15	98.70	44.74	**59.38**	68.15	3.33
		+ ∞-RoPE (2025)	97.27	96.19	98.68	48.18	58.69	67.99	3.83
		+ Deep Forcing (2025)	97.52	96.43	**98.82**	41.46	59.00	67.61	3.00
		+ LongLive-RAG (2026)	**97.53**	96.39	98.77	44.84	59.24	**68.42**	**2.00**
		+ **DensityKV**	97.39	**96.44**	98.61	**52.14**	58.15	68.38	2.83
	Causal-Forcing (2026)	Native	94.60	94.68	96.56	73.96	54.58	65.53	4.00
		+ ∞-RoPE (2025)	93.93	94.11	96.21	**90.83**	55.42	68.26	3.17
		+ Deep Forcing (2025)	93.52	93.86	95.84	84.79	55.03	66.07	4.17
		+ LongLive-RAG (2026)	95.43	94.79	**97.16**	82.29	**57.31**	**70.07**	**1.67**
		+ **DensityKV**	**95.79**	**95.46**	97.03	81.09	57.07	69.08	2.00
60s	Self-Forcing (2025)	Native	95.84	95.27	98.20	**51.72**	56.05	62.22	4.17
		+ ∞-RoPE (2025)	97.24	96.24	98.58	46.64	56.09	63.28	3.00
		+ Deep Forcing (2025)	96.08	95.38	98.24	41.44	56.68	60.81	4.17
		+ LongLive-RAG (2026)	97.60	96.51	98.70	44.69	57.19	64.97	2.17
		+ **DensityKV**	**97.67**	**96.62**	**98.75**	43.80	**57.38**	**66.68**	**1.50**
	LongLive (2025)	Native	97.13	95.89	98.61	44.56	58.17	67.56	3.33
		+ ∞-RoPE (2025)	97.00	95.85	98.53	53.36	57.48	66.94	4.25
		+ Deep Forcing (2025)	97.17	96.04	**98.73**	45.13	57.48	67.27	2.92
		+ LongLive-RAG (2026)	**97.32**	96.08	98.62	49.90	**58.30**	67.79	**1.83**
		+ **DensityKV**	97.12	**96.30**	98.56	**55.96**	57.40	**67.91**	2.67
	Causal-Forcing (2026)	Native	93.52	94.12	95.74	72.32	51.24	62.30	4.83
		+ ∞-RoPE (2025)	93.81	93.78	96.09	**92.47**	54.42	67.50	3.33
		+ Deep Forcing (2025)	94.27	94.18	96.62	78.59	52.12	64.25	3.33
		+ LongLive-RAG (2026)	94.29	94.24	96.48	88.20	54.95	68.16	2.17
		+ **DensityKV**	**95.58**	**95.30**	**96.96**	81.54	**56.40**	**68.98**	**1.33**
120s	Self-Forcing (2025)	Native	96.12	95.32	98.27	43.39	55.64	61.57	4.33
		+ ∞-RoPE (2025)	97.15	96.09	98.55	**46.29**	55.11	61.81	3.17
		+ Deep Forcing (2025)	96.92	**96.83**	**98.97**	15.23	52.84	57.93	3.50
		+ LongLive-RAG (2026)	**97.64**	96.40	98.75	44.10	56.30	64.16	2.33
		+ **DensityKV**	97.57	96.55	98.76	44.22	**56.91**	**66.55**	**1.67**
	LongLive (2025)	Native	96.93	95.64	98.58	47.12	**57.90**	66.95	3.08
		+ ∞-RoPE (2025)	96.81	95.65	98.48	53.59	56.73	66.19	4.17
		+ Deep Forcing (2025)	97.17	95.72	98.57	46.03	56.98	66.03	3.67
		+ LongLive-RAG (2026)	**97.22**	95.88	**98.62**	50.25	57.57	66.95	**1.92**
		+ **DensityKV**	96.98	**96.21**	98.51	**58.40**	57.02	**67.89**	2.17
	Causal-Forcing (2026)	Native	92.98	94.66	95.41	63.79	47.31	58.23	4.33
		+ ∞-RoPE (2025)	93.45	93.56	95.95	**93.44**	53.47	66.81	3.00
		+ Deep Forcing (2025)	93.19	94.04	95.60	76.45	46.86	61.58	4.17
		+ LongLive-RAG (2026)	94.38	94.08	96.56	90.21	54.82	68.23	2.17
		+ **DensityKV**	**95.35**	**95.19**	**96.88**	82.38	**55.99**	**68.77**	**1.33**

Table 1: Comparison across backbones. Bold/underline mark best/second-best values; Avg. Rank is averaged over the six metrics.

4 Experiments

Implementation Details

We build on the open Wan2.1-T2V-1.3B model (Wan Team et al. 2025) and use the released frozen checkpoints for LongLive, Self-Forcing, and Causal-Forcing. DensityKV requires no model fine-tuning. For quality comparisons, we match methods by the upper bound on attention-visible nonlocal K/V tokens rather than by instantaneous occupancy. Each method receives a nonlocal budget equivalent to six complete frames per head, in addition to one explicit first-frame sink and a five-frame local window. DensityKV therefore stores at most 9,360 historical KV states per attention head. Unless stated otherwise, we use $\epsilon = 1$, $p = 2$, $\sigma = 8$, $\delta = 10^{-6}$, and density-growth threshold $\tau = 2.0$. The bank is reset for each video.

Persistent temporal-state storage. We compare the aggregate persistent temporal state retained on CPU and GPU, excluding model parameters and layer-local temporary workspace for every method. LongLive-RAG retains the raw BF16 K/V states of all latent frames together with their retrieval descriptors, requiring 32.1, 64.3, and 128.5 GiB at 30, 60, and 120 seconds, respectively. Deep Forcing retains a 12-frame K/V cache, a four-frame query buffer, and metadata, totaling 3.76 GiB. At $M = 9{,}360$, DensityKV's post-RoPE K/V bank, density baselines, and provenance occupy 1.67 GiB; including its first-frame sink and

Parameter	Setting	Subject↑	Background↑	Smoothness↑	Dynamic↑	Aesthetic↑	Imaging↑	Avg. Rank↓
(a) Parameter sensitivity								
τ	1.5	97.03	96.29	98.26	**52.57**	**57.97**	68.70	2.17
	2.0*	97.19	**96.34**	98.33	51.94	57.74	**68.73**	**1.83**
	3.0	**97.20**	96.33	**98.35**	44.44	57.83	65.50	2.00
M	6,240	**97.25**	96.31	**98.34**	49.38	57.61	67.06	2.17
	9,360*	97.19	**96.34**	98.33	**51.94**	57.74	68.73	**1.67**
	12,480	97.12	96.30	98.33	51.39	**57.85**	**69.21**	2.00
σ	4	97.08	96.28	98.28	**52.64**	57.78	68.85	2.17
	8*	**97.19**	**96.34**	**98.33**	51.94	57.74	68.73	**1.83**
	16	97.18	96.27	98.32	50.49	**57.94**	**69.27**	2.00
p	1	**97.19**	96.32	98.32	50.35	57.71	**69.02**	1.83
	2*	**97.19**	**96.34**	**98.33**	**51.94**	**57.74**	68.73	**1.17**
	4	97.06	96.27	98.22	51.88	57.68	68.52	2.83
(b) Policy ablations								
Default*		**97.19**	**96.34**	98.33	51.94	57.74	68.73	**2.83**
Refresh density baselines		**97.19**	96.30	**98.42**	47.29	56.84	64.85	4.33
Source-order candidates		97.06	96.30	98.28	51.32	57.36	68.54	5.17
Densest-only eviction		97.12	96.28	98.34	51.53	57.85	69.22	3.17
Mandatory + source eviction		97.12	96.28	98.34	51.11	57.86	**69.23**	3.17
Independent per-head		97.14	96.31	98.30	50.00	**58.14**	69.05	3.33
Within-block competition		96.96	96.16	98.10	**57.50**	57.69	68.83	5.17
Geometry	pre-RoPE K	96.93	96.18	98.23	51.88	**57.38**	**67.79**	2.33
	pre-RoPE $[K;V]$	96.93	96.22	**98.28**	**52.64**	57.24	66.41	2.00
	post-RoPE K	**97.01**	**96.23**	**98.28**	49.65	57.31	67.01	**1.83**
	post-RoPE $[K;V]$	96.88	96.20	**98.28**	50.56	57.17	65.91	3.17

Table 2: Parameter sensitivity and policy ablations. Bold/underline mark best/second-best displayed values within each ablation group.

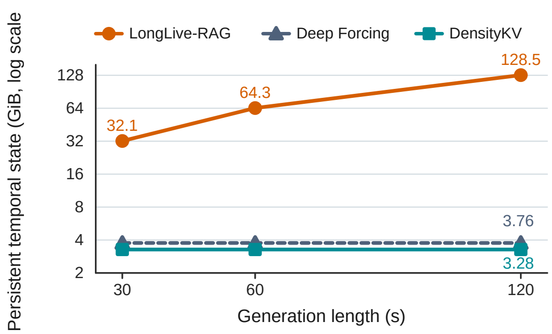

Figure 4: Persistent temporal-state storage versus generation length. LongLive-RAG grows linearly, whereas Deep Forcing and DensityKV remain bounded.

five-frame local K/V window yields 3.28 GiB of total persistent temporal-state storage. DensityKV thus reduces storage relative to LongLive-RAG by $9.8\times$, $19.6\times$, and $39.2\times$ at the same horizons (Figure 4).

General Setup

Our main evaluation follows LongLive-RAG's long-video protocol (Hu et al. 2026), using all 128 MovieGenBench prompts refined with Qwen2.5-7B-Instruct. We compare Native, ∞-RoPE, Deep Forcing, LongLive-RAG, and DensityKV on LongLive, Self-Forcing, and Causal-Forcing. We use the baseline scores reported by LongLive-RAG under this protocol and evaluate DensityKV with matched released checkpoints, prompts, sampler settings, and seeds.

Controlled ablations use a fixed 16-prompt subset from the same benchmark. Parameter sensitivity sweeps τ, bank capacity M, σ, and p around the default $(2.0, 9{,}360, 8, 2)$, with $\epsilon = 1$ fixed by the scale normalization above; discrete ablations change exactly one policy component.

Metrics

VBench defines 16 dimensions (Huang et al. 2023); following LongLive-RAG, we report the six used by its long-video protocol: Subject Consistency, Background Consistency, Motion Smoothness, Dynamic Degree, Aesthetic Quality, and Imaging Quality. Higher is better for all six. Avg. Rank averages method ranks across these metrics (lower is better); three-backbone ablations report unweighted means across backbones.

Main Results

Comparison across backbones.

Figure 3 compares the native backbone, LongLive-RAG, and DensityKV on two matched cases. DensityKV better preserves identity, hairstyle, sunglasses, and street structure on LongLive. On Causal-Forcing, the native model develops late subject and scene distortions; LongLive-RAG mitigates them, while DensityKV preserves subject appearance at 60 seconds. These cases qualitatively reflect the aggregate long-horizon gains.

Table 1 compares five methods on three backbones, three horizons, and the six selected VBench metrics. Figure 5 summarizes single-metric wins: DensityKV leads at every horizon with 8, 13, and 10 wins at 30, 60, and 120 seconds. Overall, it obtains 31 of 54 wins and the best Avg. Rank of 1.89, while improving Background Consistency over LongLive-RAG in all nine horizon–backbone comparisons.

Gains vary by backbone. DensityKV ranks first at all Self-Forcing horizons. On Causal-Forcing it reaches an Avg. Rank of 1.33 at 60 and 120 seconds and achieves the best scores on five metrics, while ∞-RoPE achieves a higher Dynamic

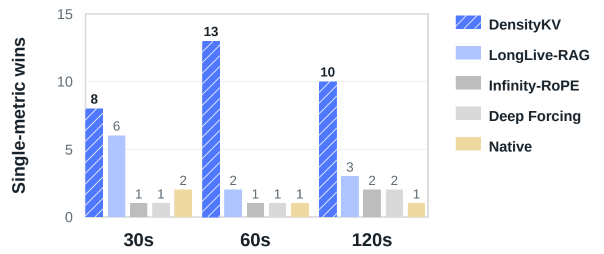

Figure 5: Single-metric wins under the matched LongLive-RAG protocol at 30, 60, and 120 seconds.

Degree. This trade-off suggests that stronger historical anchoring can reduce motion diversity on this high-motion backbone. On LongLive, DensityKV achieves the best Background Consistency, Dynamic Degree, and Imaging Quality at 60 and 120 seconds, whereas LongLive-RAG retains stronger Subject Consistency and Aesthetic Quality. Overall, DensityKV transfers across all three backbones, with gains shaped by each generator's error profile.

Ablation Study

Parameter sensitivity. Table 2(a) varies the density-growth threshold, capacity, and kernel shape around $(\tau, M, \sigma, p) = (2.0, 9{,}360, 8, 2)$ with $\epsilon = 1$ fixed. Avg. Rank is computed over the six displayed macro-averaged metrics within each sweep. At 60 seconds, $\tau = 2.0$ ranks best at 1.83, while $\tau = 1.5$ favors dynamics and aesthetics and $\tau = 3.0$ favors consistency. Performance varies non-monotonically with capacity: $M = 9{,}360$ ranks best at 1.67, balancing the stronger subject consistency of $M = 6{,}240$ with the visual quality of $M = 12{,}480$. Increasing σ to 16 improves aesthetic and imaging quality, whereas $\sigma = 8$ improves subject, background, smoothness, and dynamics and obtains the best Avg. Rank of 1.83. The default $p = 2$ ranks best in its sweep at 1.17. We therefore use $(2.0, 9{,}360, 8, 2)$ as the balanced default across consistency, dynamics, and visual quality, with $\epsilon = 1$ serving only as the fixed kernel normalization.

Policy ablations. Table 2(b) isolates candidate ordering, density baselines, eviction selection, cross-head synchronization, within-block competition, and retention geometry.

Candidate ordering has the largest effect on the bank's coverage structure. Removing density guidance and processing candidates in source order gives an Avg. Rank of 5.17, compared with 2.83 for the density-guided default. Averaging the normalized increment prevents a single sensitive anchor from dominating the candidate order and better prioritizes states that improve coverage across the bank.

Freezing the density baseline recorded at admission is important for preventing the bank from following the current generation drift. Refreshing the baselines after every update attains the highest Motion Smoothness of 98.42, but reduces Imaging Quality to 64.85 and gives an Avg. Rank of 4.33. If the reference density continually adapts to the current bank, increasingly crowded regions can repeatedly redefine their own normal state, weakening the historical constraint.

The eviction variants are more stable than the admission variants. Densest-only eviction and source-ordered auxiliary eviction both obtain an Avg. Rank of 3.17, but the former better preserves dynamics while the latter favors aesthetics and imaging quality. This indicates that eviction selection primarily controls the trade-off among these criteria.

Allowing each attention head to choose its admission count independently raises Aesthetic Quality from 57.74 to 58.14 and Imaging Quality from 68.73 to 69.05, but slightly lowers Subject Consistency, Background Consistency, Motion Smoothness, and Dynamic Degree, yielding an Avg. Rank of 3.33. Sharing the admission count therefore provides a more balanced update pace across heads.

The retention-geometry ablation supports measuring redundancy with post-RoPE keys alone. Post-RoPE K obtains the best Avg. Rank of 1.83, achieving the highest Subject and Background Consistency and tying for the best Motion Smoothness. Its lower Dynamic Degree (49.65) reflects a stability–dynamics trade-off: stronger attention-aligned historical anchoring improves long-horizon consistency while reducing motion diversity. Pre-RoPE K and $[K; V]$ obtain Avg. Ranks of 2.33 and 2.00, respectively, while post-RoPE $[K; V]$ obtains the worst Avg. Rank of 3.17. This result is consistent with the role of keys in attention: queries address historical states through post-RoPE key similarity, whereas values determine the returned content but do not define the query-to-memory correspondence. We therefore use post-RoPE K alone for all density decisions.

Immediate within-block competition directly tests insertion hysteresis by performing an additional density cleanup after committing each candidate block, so co-admitted states compete before their baselines are frozen. It raises Dynamic Degree to 57.50, but gives the lowest Subject Consistency, Background Consistency, and Motion Smoothness and an Avg. Rank of 5.17. This result suggests that immediate absolute-density balancing can remove coherent within-block evidence; delaying this competition preserves local continuity, while subsequent blocks still trigger post-admission density-growth eviction.

Overall, density-guided candidate ordering and frozen insertion baselines are the most consequential policy components. Eviction and cross-head synchronization primarily control the trade-off among consistency, dynamics, and visual quality. The default obtains the best Avg. Rank among update-policy variants (2.83) and is therefore used in the main experiments.

5 Conclusion

DensityKV is a training-free historical KV selection algorithm that controls post-admission crowding in post-RoPE key space while preserving exact K/V pairs. Across three frozen video backbones, it improves long-horizon consistency under the same historical-capacity bound and keeps storage independent of rollout length. By allocating history at token and head granularity, the method preserves non-local subject and scene evidence without training or modifying the generator. Because retained values are neither merged nor reconstructed, every selected state remains directly usable by the backbone's native attention.

References

Cai, Z.; Xiao, W.; Sun, H.; Luo, C.; Zhang, Y.; Wan, K.; Li, Y.; Zhou, Y.; Chang, L.-W.; Gu, J.; Dong, Z.; Anandkumar, A.; Asi, A.; and Hu, J. 2025. R-KV: Redundancy-aware KV Cache Compression for Reasoning Models. In *Advances in Neural Information Processing Systems*.

Cai, Z.; Zhang, Y.; Gao, B.; Liu, Y.; Liu, T.; Lu, K.; Xiong, W.; Dong, Y.; Chang, B.; Hu, J.; and Xiao, W. 2024. PyramidKV: Dynamic KV Cache Compression based on Pyramidal Information Funneling. *arXiv preprint arXiv:2406.02069*.

Chen, H.; Xu, C.; Yang, X.; Chen, X.; and Deng, C. 2026. Past- and Future-Informed KV Cache Policy with Salience Estimation in Autoregressive Video Diffusion. *arXiv preprint arXiv:2601.21896*.

Hardin, D. P.; and Saff, E. B. 2005. Minimal Riesz Energy Point Configurations for Rectifiable d-Dimensional Manifolds. *Advances in Mathematics*, 193(1): 174–204.

Hu, J.; Wang, S.; He, Y.; Gong, P.; Yi, J.; Zhang, J.; Bai, Y.; Chen, R.; Zhang, G.; Li, C.; and Yuan, K. 2025. CentroidKV: Efficient Long-Context LLM Inference via KV Cache Clustering. *arXiv preprint arXiv:2506.11418*.

Hu, Q.; Yang, S.; Huang, W.; Han, S.; and Chen, Y. 2026. LongLive-RAG: A General Retrieval-Augmented Framework for Long Video Generation. *arXiv preprint arXiv:2606.02553*.

Huang, X.; Li, Z.; He, G.; Zhou, M.; and Shechtman, E. 2025. Self Forcing: Bridging the Train-Test Gap in Autoregressive Video Diffusion. *arXiv preprint arXiv:2506.08009*.

Huang, Z.; He, Y.; Yu, J.; Zhang, F.; Si, C.; Jiang, Y.; Zhang, Y.; Wu, T.; Jin, Q.; Chanpaisit, N.; et al. 2023. VBench: Comprehensive Benchmark Suite for Video Generative Models. *arXiv preprint arXiv:2311.17982*.

Ji, Y.; Zhong, Z.; Zhang, J.; Yang, Q.; Jin, X.; Qin, Y.; Luo, W.; Mao, S.; Liu, W.; and Li, H. 2026. Forcing-KV: Hybrid KV Cache Compression for Efficient Autoregressive Video Diffusion Models. *arXiv preprint arXiv:2605.09681*.

Kim, J.; Kang, J.; Choi, J.; and Han, B. 2024. FIFO-Diffusion: Generating Infinite Videos from Text without Training. *arXiv preprint arXiv:2405.11473*.

Li, Y.; Huang, Y.; Yang, B.; Venkitesh, B.; Locatelli, A.; Ye, H.; Cai, T.; Lewis, P.; and Chen, D. 2024. SnapKV: LLM Knows What You Are Looking for Before Generation. *Advances in Neural Information Processing Systems*, 37.

Liu, G.; Li, C.; Zhao, J.; Zhang, C.; and Guo, M. 2025. ClusterKV: Manipulating LLM KV Cache in Semantic Space for Recallable Compression. In *Proceedings of the 62nd ACM/IEEE Design Automation Conference*, 1–7.

Luo, J.; Liu, Q.; Wang, T.; Liu, J.; Chen, J.; Wang, C.; Zhu, H.; Gao, C.; Hu, X.; Sun, Q.; and Chen, Z. 2026. Future Forcing: Future-aware Training-free KV Cache Policy for Autoregressive Video Generation. *arXiv preprint arXiv:2605.30083*.

Meng, Y.; Luo, X.; Li, L.; Jiang, W.; Gao, C.; Chen, X.; Li, Y.; and Zhang, X.-P. 2026. TetherCache: Stabilizing Autoregressive Long-Form Video Generation with Gated Recall and Trusted Alignment. *arXiv preprint arXiv:2606.13035*.

Peebles, W.; and Xie, S. 2023. Scalable Diffusion Models with Transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 4195–4205.

Qiu, H.; Xia, M.; Zhang, Y.; He, Y.; Wang, X.; Shan, Y.; and Liu, Z. 2023. FreeNoise: Tuning-Free Longer Video Diffusion via Noise Rescheduling. *arXiv preprint arXiv:2310.15169*.

Wan Team; Wang, A.; Ai, B.; Wen, B.; Mao, C.; Xie, C.-W.; Chen, D.; Yu, F.; Zhao, H.; Yang, J.; et al. 2025. Wan: Open and Advanced Large-Scale Video Generative Models. *arXiv preprint arXiv:2503.20314*.

Wu, M.; Feng, W.; Zhang, Z.; Qin, H.; Li, Y.; Fan, G.; Liu, X.; An, Z.; Huang, L.; Xu, Y.; and Yang, C. 2026. Echo-Forcing: A Scene Memory Framework for Interactive Long Video Generation. *arXiv preprint arXiv:2605.16003*.

Xiao, G.; Tian, Y.; Chen, B.; Han, S.; and Lewis, M. 2023. Efficient Streaming Language Models with Attention Sinks. *arXiv preprint arXiv:2309.17453*.

Yang, S.; Huang, W.; Chu, R.; Xiao, Y.; Zhao, Y.; Wang, X.; Li, M.; Xie, E.; Chen, Y.; Lu, Y.; Han, S.; and Chen, Y. 2025. LongLive: Real-time Interactive Long Video Generation. *arXiv preprint arXiv:2509.22622*.

Ye, B.; Cui, X.; Zhao, J.; Wei, T.; and Zhang, M.-L. 2026. DySink: Dynamic Frame Sinks for Autoregressive Long Video Generation. *arXiv preprint arXiv:2605.21028*.

Yesiltepe, H.; Hu, J.; Meral, T. H. S.; Akan, A. K.; Oktay, K.; Eldardiry, H.; and Yanardag, P. 2026. VideoMLA: Low-Rank Latent KV Cache for Minute-Scale Autoregressive Video Diffusion. *arXiv preprint arXiv:2605.30351*.

Yesiltepe, H.; Meral, T. H. S.; Akan, A. K.; Oktay, K.; and Yanardag, P. 2025. Infinity-RoPE: Action-Controllable Infinite Video Generation Emerges From Autoregressive Self-Rollout. *arXiv preprint arXiv:2511.20649*.

Yi, J.; Jang, W.; Cho, P. H.; Nam, J.; Yoon, H.; and Kim, S. 2025. Deep Forcing: Training-Free Long Video Generation with Deep Sink and Participative Compression. *arXiv preprint arXiv:2512.05081*.

Yin, T.; Zhang, Q.; Zhang, R.; Freeman, W. T.; Durand, F.; Shechtman, E.; and Huang, X. 2024. From Slow Bidirectional to Fast Autoregressive Video Diffusion Models. *arXiv preprint arXiv:2412.07772*.

Zandieh, A.; Han, I.; Mirrokni, V.; and Karbasi, A. 2024. SubGen: Token Generation in Sublinear Time and Memory. *arXiv preprint arXiv:2402.06082*.

Zhang, Z.; Sheng, Y.; Zhou, T.; Chen, T.; Zheng, L.; Cai, R.; Song, Z.; Tian, Y.; Ré, C.; Barrett, C.; Wang, Z.; and Chen, B. 2023. H_2O: Heavy-Hitter Oracle for Efficient Generative Inference of Large Language Models. *Advances in Neural Information Processing Systems*, 36.

Zhao, Z.; Lu, Y.; Liu, Z.; Song, J.; Deng, J.; and Patras, I. 2026. Relax Forcing: Relaxed KV-Memory for Consistent Long Video Generation. *arXiv preprint arXiv:2603.21366*.

Zhu, H.; Zhao, M.; He, G.; Su, H.; Li, C.; and Zhu, J. 2026. Causal Forcing: Autoregressive Diffusion Distillation Done Right for High-Quality Real-Time Interactive Video Generation. *arXiv preprint arXiv:2602.02214*.